

# 1 Quantitative Approximation by the Mean-Field Limit for Low-Rank Networks

This section provides a rigorous quantitative bound on the approximation error between finite-width low-rank networks and their mean-field limit. The key result shows that the approximation error scales as  $O(1/\sqrt{n_{\min}} + \sqrt{\epsilon})$  with explicit constants, where  $n_{\min}$  is the minimum width across layers and  $\epsilon$  is the learning rate step size. This bound holds for any  $n_1$  and  $n_2$ , independent of the data dimension  $d$ , similar to the full-width case [?, ?]. The bound suggests that widths  $n_1, n_2 \approx 1000$  are typically sufficient to observe mean-field behaviors, as empirically validated in [?] for high-dimensional real-world data.

## 1.1 Main Result: Approximation by the MF Limit

**Assumption 1.1** (Initialization for Low-Rank Networks). We assume that  $\text{ess-sup} \max_{1 \leq k \leq r} |w_1^0(C_1, k)| \leq K$  and  $\text{ess-sup} |w_2^0(C_2)| \leq K$ , where  $w_1^0$  and  $w_2^0$  are the initial weights as described in the low-rank architecture setup.

**Theorem 1.2** (Quantitative Approximation for Low-Rank Networks). *Given a family  $\text{Init}$  of initialization laws and a tuple  $\{n_1, n_2\}$  that is in the index set of  $\text{Init}$ , perform the coupling procedure for the low-rank architecture as described in Section ?? . Fix a terminal time  $T \in \epsilon \mathbb{N}_{\geq 0}$ . Under Assumptions ??, 1.1, and the low-rank structure with mixing matrix  $L$  satisfying  $\|L\|_{\infty, 1} \leq rK$ , for  $\epsilon \leq 1$ , we have with probability at least  $1 - 2\delta$ ,*

$$\begin{aligned} \mathcal{D}_T(W, \mathbf{W}) &\leq e^{K_T(1+rK)} \left( \frac{1}{\sqrt{n_{\min}}} + \sqrt{\epsilon} \right) \log^{1/2} \left( \frac{3(T+1)n_{\max}^2}{\delta} + e \right) \\ &\equiv \text{err}_{\delta, T}(\epsilon, n_1, n_2, r), \end{aligned}$$

in which  $n_{\min} = \min\{n_1, n_2\}$ ,  $n_{\max} = \max\{n_1, n_2\}$ ,  $K_T = K(1 + T^K)$ , and the factor  $(1 + rK)$  accounts for the low-rank structure through  $\|L\|_{\infty, 1} \leq rK$ .

The theorem gives a connection between  $\mathbf{W}(\lfloor t/\epsilon \rfloor)$  (the discrete-time finite-width network) and  $W(t)$  (the continuous-time mean-field limit). This result extends the quantitative approximation theorem from the full-width case [?] to the low-rank setting, with the key difference being the multiplicative factor  $(1 + rK)$  in the exponential constant, which accounts for the low-rank mixing matrix structure.

**Corollary 1.3** (Approximation Quality for Test Functions). *Under the same setting as Theorem 1.1, consider any test function  $\psi : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  which is  $K$ -Lipschitz in the second variable uniformly in the first variable (an example of  $\psi$  is the loss  $\mathcal{L}$ ). For any  $\delta > 0$ , with probability at least  $1 - 3\delta$ ,*

$$\sup_{t \leq T} |\mathbb{E}_Z [\psi(Y, \hat{y}(X; \mathbf{W}(\lfloor t/\epsilon \rfloor)))] - \mathbb{E}_Z [\psi(Y, \hat{y}(X; W(t)))]| \leq e^{K_T(1+rK)} \text{err}_{\delta, T}(\epsilon, n_1, n_2, r),$$

where  $\hat{y}(X; \mathbf{W})$  and  $\hat{y}(X; W)$  denote the outputs of the finite-width and mean-field networks respectively.

*Proof sketch.* The proof follows the same structure as in the full-width case [?]. Since  $\psi$  is  $K$ -Lipschitz in the second variable, we have:

$$|\psi(Y, \hat{\mathbf{y}}(X; \mathbf{W})) - \psi(Y, \hat{y}(X; W))| \leq K |\hat{\mathbf{y}}(X; \mathbf{W}) - \hat{y}(X; W)|.$$

The difference  $|\hat{\mathbf{y}}(X; \mathbf{W}) - \hat{y}(X; W)|$  can be bounded by the distance  $\mathcal{D}_T(W, \mathbf{W})$  using the low-rank structure of the network. Specifically, for the low-rank architecture, the output difference involves:

$$\begin{aligned} |\hat{\mathbf{y}}(X; \mathbf{W}) - \hat{y}(X; W)| &\leq \sum_{k=1}^r |L_{C_2, k}| \left| \frac{1}{n_1} \sum_{j_1=1}^{n_1} \mathbf{w}_1(\lfloor t/\epsilon \rfloor, j_1, k) \varphi_1(\langle f_1(C_1(j_1)), X \rangle) \right. \\ &\quad \left. - \mathbb{E}_{C_1}[w_1(t, C_1, k) \varphi_1(\langle f_1(C_1), X \rangle)] \right| + (\text{similar terms for } w_2) \\ &\leq rK \mathcal{D}_T(W, \mathbf{W}) + K \mathcal{D}_T(W, \mathbf{W}) = K(1+rK) \mathcal{D}_T(W, \mathbf{W}), \end{aligned}$$

where we used  $\|L\|_{\infty, 1} \leq rK$ . Taking expectation over  $Z$  and applying Theorem 1.1 completes the proof.  $\square$

**Comparison with the Full-Width Case.** The main difference between our low-rank result and the full-width case [?] is the appearance of the factor  $(1+rK)$  in the exponential constant. In the full-width case, the bound is:

$$\mathcal{D}_T(W, \mathbf{W}) \leq e^{K_T} \left( \frac{1}{\sqrt{n_{\min}}} + \sqrt{\epsilon} \right) \log^{1/2} \left( \frac{3(T+1)n_{\max}^2}{\delta} + e \right),$$

whereas in our low-rank case, we have:

$$\mathcal{D}_T(W, \mathbf{W}) \leq e^{K_T(1+rK)} \left( \frac{1}{\sqrt{n_{\min}}} + \sqrt{\epsilon} \right) \log^{1/2} \left( \frac{3(T+1)n_{\max}^2}{\delta} + e \right).$$

The factor  $(1+rK)$  arises from:

- The low-rank structure of  $H_2$ , which involves a sum over  $r$  channels:  $H_2(t, c_2; X, W) = \sum_{k=1}^r L_{c_2, k} m_k(t; X, W)$ .
- The mixing matrix bounds:  $\|L\|_{\infty, 1} = \sup_{c_2} \sum_{k=1}^r |L_{c_2, k}| \leq rK$ .
- The backpropagated signal  $B_k$  which aggregates over  $n_2$  neurons with mixing coefficients  $L_{C_2, k}$ .

However, the fundamental structure of the proof remains the same: we still decompose the error into particle coupling and gradient descent discretization, and the scaling with  $n_{\min}$  and  $\epsilon$  is identical. The low-rank structure only affects the exponential constant, not the polynomial scaling. This suggests that the mean-field approximation quality is preserved under low-rank constraints, with the trade-off being a potentially larger (but still finite) exponential constant.

## 1.2 Particle ODEs for Low-Rank Networks

We construct auxiliary trajectories, which we call the *particle ODEs* for the low-rank case. These are continuous-time trajectories of finitely many neurons, averaged over the data distribution, adapted to the low-rank structure:

$$\begin{aligned}\frac{\partial}{\partial t} \tilde{w}_2(t, j_2) &= -\xi_2(t) \mathbb{E}_Z \left[ d_L(Z; \tilde{W}(t)) \varphi_2(\tilde{H}_2(t, j_2; X, \tilde{W}(t))) \right], \\ \frac{\partial}{\partial t} \tilde{w}_1(t, j_1, k) &= -\xi_1(t) \mathbb{E}_Z \left[ d_L(Z; \tilde{W}(t)) \varphi_1(\langle f_1(C_1(j_1)), X \rangle) \tilde{B}_k(t; X, \tilde{W}(t)) \right],\end{aligned}$$

where  $j_1 = 1, \dots, n_1$ ,  $j_2 = 1, \dots, n_2$ ,  $k = 1, \dots, r$ ,  $\tilde{W}(t) = (\tilde{w}_1(t, \cdot, \cdot), \tilde{w}_2(t, \cdot))$ , and  $t \in \mathbb{R}_{\geq 0}$ .

The second-layer output for the particle ODEs is:

$$\tilde{H}_2(t, j_2; X, \tilde{W}) = \sum_{k=1}^r L_{C_2(j_2), k} \frac{1}{n_1} \sum_{j_1=1}^{n_1} \tilde{w}_1(t, j_1, k) \varphi_1(\langle f_1(C_1(j_1)), X \rangle),$$

and the backpropagated signal is:

$$\tilde{B}_k(t; X, \tilde{W}) = \frac{1}{n_2} \sum_{j_2=1}^{n_2} L_{C_2(j_2), k} \varphi_2'(\tilde{H}_2(t, j_2; X, \tilde{W})) \tilde{w}_2(t, j_2).$$

We specify the initialization  $\tilde{W}(0)$ :  $\tilde{w}_1(0, j_1, k) = w_1^0(C_1(j_1), k)$  and  $\tilde{w}_2(0, j_2) = w_2^0(C_2(j_2))$ . That is, it shares the same initialization with the neural network  $\mathbf{W}(0)$ , and hence is coupled with the neural network and the MF ODEs.

The existence and uniqueness of the solution to the particle ODEs follows from the same proof as in Theorem ??, adapted to account for the low-rank structure. We equip  $\tilde{W}(t)$  with the norm:

$$\|\tilde{W}\|_T = \max \left\{ \max_{j_1 \leq n_1, 1 \leq k \leq r} \sup_{t \leq T} |\tilde{w}_1(t, j_1, k)|, \max_{j_2 \leq n_2} \sup_{t \leq T} |\tilde{w}_2(t, j_2)| \right\}.$$

We define the distance measures:

$$\begin{aligned}\mathcal{D}_T(W, \tilde{W}) &= \sup \left\{ |w_1(t, C_1(j_1), k) - \tilde{w}_1(t, C_1(j_1), k)|, |w_2(t, C_2(j_2)) - \tilde{w}_2(t, C_2(j_2))| : \right. \\ &\quad \left. t \leq T, j_1 \leq n_1, j_2 \leq n_2, 1 \leq k \leq r \right\}, \\ \mathcal{D}_T(\tilde{W}, \mathbf{W}) &= \sup \left\{ |\mathbf{w}_1(\lfloor t/\epsilon \rfloor, j_1, k) - \tilde{w}_1(t, C_1(j_1), k)|, \right. \\ &\quad \left. |\mathbf{w}_2(\lfloor t/\epsilon \rfloor, j_2) - \tilde{w}_2(t, C_2(j_2))| : t \leq T, j_1 \leq n_1, j_2 \leq n_2, 1 \leq k \leq r \right\}.\end{aligned}$$

## 1.3 Key Lemmas

**Lemma 1.4** (Particle Coupling for Low-Rank Networks). *Under the same setting as Theorem 1.1, for any  $\delta > 0$ , with probability at least  $1 - \delta$ ,*

$$\mathcal{D}_T(W, \tilde{W}) \leq \frac{1}{\sqrt{n_{\min}}} \log^{1/2} \left( \frac{3(T+1)n_{\max}^2}{\delta} + e \right) e^{K_T(1+rK)},$$

in which  $n_{\min} = \min\{n_1, n_2\}$ ,  $n_{\max} = \max\{n_1, n_2\}$ , and  $K_T = K(1 + T^K)$ .

**Lemma 1.5** (Gradient Descent Discretization for Low-Rank Networks). *Under the same setting as Theorem 1.1, for any  $\delta > 0$  and  $\epsilon \leq 1$ , with probability at least  $1 - \delta$ ,*

$$\mathcal{D}_T(\tilde{W}, \mathbf{W}) \leq \sqrt{\epsilon \log \left( \frac{2n_1 n_2 r}{\delta} + e \right)} e^{K_T(1+rK)},$$

in which  $K_T = K(1 + T^K)$ .

*Proof of Theorem 1.1.* Using the triangle inequality:

$$\mathcal{D}_T(W, \mathbf{W}) \leq \mathcal{D}_T(W, \tilde{W}) + \mathcal{D}_T(\tilde{W}, \mathbf{W}),$$

the result follows immediately from Lemmas 1.3 and 1.3, noting that the  $\log(2n_1 n_2 r / \delta)$  term can be absorbed into the  $\log(3(T+1)n_{\max}^2 / \delta + e)$  term up to constants.  $\square$

## 1.4 Proof of Lemma 1.3

*Proof.* In the following, let  $K_t$  denote a generic positive constant that may change from line to line and takes the form  $K_t = K(1 + t^K)$ , such that  $K_t \geq 1$  and  $K_t \leq K_T$  for all  $t \leq T$ . We first note that at initialization,  $\mathcal{D}_0(W, \tilde{W}) = 0$ . Since  $\llbracket W \rrbracket 0 \leq K$ , we have  $\llbracket W \rrbracket T \leq K_T$  by the a priori bounds. Furthermore, it is easy to see that  $\llbracket \tilde{W} \rrbracket 0 \leq \llbracket W \rrbracket 0 \leq K$  almost surely. By the same argument,  $\llbracket \tilde{W} \rrbracket T \leq K_T$  almost surely.

We decompose the proof into several steps.

**Step 1: Main bound with low-rank structure.** Let us define, for brevity, the differences specific to the low-rank architecture:

$$\begin{aligned} q_2(t, x, j_2, c_2) &= \tilde{H}_2(t, j_2; x, \tilde{W}(t)) - H_2(t, c_2; x, W(t)), \\ q_{B,k}(t, x) &= \tilde{B}_k(t; x, \tilde{W}(t)) - B_k(t; x, W(t)), \end{aligned}$$

where  $H_2(t, c_2; x, W) = \sum_{k=1}^r L_{c_2,k} \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle f_1(C_1), x \rangle)]$  and  $B_k(t; x, W) = \mathbb{E}_{C_2} [L_{C_2,k} \varphi_2'(H_2(t, C_2; x, W)) w_2(t, C_2)]$ .

Consider  $t \geq 0$ . We first bound the difference in the updates between  $W$  and  $\tilde{W}$ .

**Bound for  $w_2$  and  $\tilde{w}_2$ :** By Assumption ?? and the definition of the ODEs:

$$\begin{aligned}
& \left| \frac{\partial}{\partial t} \tilde{w}_2(t, j_2) - \frac{\partial}{\partial t} w_2(t, C_2(j_2)) \right| \\
&= \left| \xi_2(t) \mathbb{E}_Z \left[ d_L(Z; \tilde{W}(t)) \varphi_2(\tilde{H}_2(t, j_2; X, \tilde{W}(t))) - d_L(Z; W(t)) \varphi_2(H_2(t, C_2(j_2); X, W(t))) \right] \right| \\
&\leq K \mathbb{E}_Z \left[ |d_L(Z; \tilde{W}(t)) - d_L(Z; W(t))| |\varphi_2(\tilde{H}_2(t, j_2; X, \tilde{W}(t)))| \right] \\
&\quad + K \mathbb{E}_Z \left[ |d_L(Z; W(t))| |\varphi_2(\tilde{H}_2(t, j_2; X, \tilde{W}(t))) - \varphi_2(H_2(t, C_2(j_2); X, W(t)))| \right] \\
&\leq K_t \mathbb{E}_Z [|q_2(t, X, j_2, C_2(j_2))|] + K_t \mathcal{D}_t(W, \tilde{W}),
\end{aligned}$$

where we used that  $d_L$  is Lipschitz in  $W$ ,  $\varphi_2$  is Lipschitz, and  $H_2$  differences are controlled.

**Bound for  $w_1$  and  $\tilde{w}_1$ :** For the low-rank case, we have  $k = 1, \dots, r$  channels. By the definition of the ODEs:

$$\begin{aligned}
& \left| \frac{\partial}{\partial t} \tilde{w}_1(t, j_1, k) - \frac{\partial}{\partial t} w_1(t, C_1(j_1), k) \right| \\
&= \left| \xi_1(t) \mathbb{E}_Z \left[ d_L(Z; \tilde{W}(t)) \varphi_1(\langle f_1(C_1(j_1)), X \rangle) \tilde{B}_k(t; X, \tilde{W}(t)) \right. \right. \\
&\quad \left. \left. - d_L(Z; W(t)) \varphi_1(\langle f_1(C_1(j_1)), X \rangle) B_k(t; X, W(t)) \right] \right| \\
&\leq K \mathbb{E}_Z \left[ |d_L(Z; \tilde{W}(t)) - d_L(Z; W(t))| |\tilde{B}_k(t; X, \tilde{W}(t))| \right] \\
&\quad + K \mathbb{E}_Z [|d_L(Z; W(t))| |q_{B,k}(t, X)|] \\
&\leq K_t \mathbb{E}_Z [|q_{B,k}(t, X)|] + K_t \mathcal{D}_t(W, \tilde{W}),
\end{aligned}$$

where the expectation over  $j_2$  in  $\tilde{B}_k$  will be handled via concentration.

**Step 2: Decomposition of  $q_2$  with low-rank structure.** The key difference from the full-width case is that  $H_2$  involves a sum over  $r$  channels. We decompose:

$$\begin{aligned}
|q_2(t, x, j_2, c_2)| &= \left| \sum_{k=1}^r L_{C_2(j_2), k} \left( \frac{1}{n_1} \sum_{j_1=1}^{n_1} \tilde{w}_1(t, j_1, k) \varphi_1(\langle f_1(C_1(j_1)), x \rangle) \right. \right. \\
&\quad \left. \left. - \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle f_1(C_1), x \rangle)] \right) \right| \\
&\leq \sum_{k=1}^r |L_{C_2(j_2), k}| \left| \frac{1}{n_1} \sum_{j_1=1}^{n_1} \tilde{w}_1(t, j_1, k) \varphi_1(\langle f_1(C_1(j_1)), x \rangle) \right. \\
&\quad \left. - \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle f_1(C_1), x \rangle)] \right|.
\end{aligned}$$

We further decompose each term in the sum:

$$\begin{aligned}
& \left| \frac{1}{n_1} \sum_{j_1=1}^{n_1} \tilde{w}_1(t, j_1, k) \varphi_1(\langle f_1(C_1(j_1)), x \rangle) - \mathbb{E}_{C_1}[w_1(t, C_1, k) \varphi_1(\langle f_1(C_1), x \rangle)] \right| \\
& \leq \max_{j_1 \leq n_1} |\tilde{w}_1(t, j_1, k) \varphi_1(\langle f_1(C_1(j_1)), x \rangle) - w_1(t, C_1(j_1), k) \varphi_1(\langle f_1(C_1(j_1)), x \rangle)| \\
& \quad + \left| \frac{1}{n_1} \sum_{j_1=1}^{n_1} w_1(t, C_1(j_1), k) \varphi_1(\langle f_1(C_1(j_1)), x \rangle) - \mathbb{E}_{C_1}[w_1(t, C_1, k) \varphi_1(\langle f_1(C_1), x \rangle)] \right| \\
& \equiv Q_{2,1,k}(x, j_2) + Q_{2,2,k}(x, j_2).
\end{aligned}$$

Therefore:

$$|q_2(t, x, j_2, c_2)| \leq \sum_{k=1}^r |L_{C_2(j_2),k}| (Q_{2,1,k}(x, j_2) + Q_{2,2,k}(x, j_2)) \leq rK \max_{1 \leq k \leq r} (Q_{2,1,k}(x, j_2) + Q_{2,2,k}(x, j_2)),$$

where we used  $\|L\|_{\infty,1} \leq rK$ .

**Step 3: Decomposition of  $q_{B,k}$  with low-rank structure.** For the back-propagated signal difference:

$$\begin{aligned}
|q_{B,k}(t, x)| &= \left| \frac{1}{n_2} \sum_{j_2=1}^{n_2} L_{C_2(j_2),k} \varphi_2'(\tilde{H}_2(t, j_2; x, \tilde{W})) \tilde{w}_2(t, j_2) \right. \\
&\quad \left. - \mathbb{E}_{C_2}[L_{C_2,k} \varphi_2'(H_2(t, C_2; x, W)) w_2(t, C_2)] \right|.
\end{aligned}$$

We decompose:

$$\begin{aligned}
|q_{B,k}(t, x)| &\leq \max_{j_2 \leq n_2} \left| L_{C_2(j_2),k} \varphi_2'(\tilde{H}_2(t, j_2; x, \tilde{W})) \tilde{w}_2(t, j_2) \right. \\
&\quad \left. - L_{C_2(j_2),k} \varphi_2'(H_2(t, C_2(j_2); x, W)) w_2(t, C_2(j_2)) \right| \\
&\quad + \left| \frac{1}{n_2} \sum_{j_2=1}^{n_2} L_{C_2(j_2),k} \varphi_2'(H_2(t, C_2(j_2); x, W)) w_2(t, C_2(j_2)) \right. \\
&\quad \left. - \mathbb{E}_{C_2}[L_{C_2,k} \varphi_2'(H_2(t, C_2; x, W)) w_2(t, C_2)] \right| \\
&\equiv Q_{B,1,k}(x) + Q_{B,2,k}(x).
\end{aligned}$$

For  $Q_{B,1,k}$ , using Assumption ?? and the fact that  $|L_{C_2(j_2),k}| \leq K$ :

$$\begin{aligned}
\mathbb{E}_Z[Q_{B,1,k}(X)] &\leq K \max_{j_2 \leq n_2} (|\tilde{w}_2(t, j_2) - w_2(t, C_2(j_2))| \\
&\quad + |w_2(t, C_2(j_2))| \mathbb{E}_Z[|\tilde{H}_2(t, j_2; X, \tilde{W}) - H_2(t, C_2(j_2); X, W)|]) \\
&\leq K_t \left( \mathcal{D}_t(W, \tilde{W}) + \max_{j_2 \leq n_2} \mathbb{E}_Z[|q_2(t, X, j_2, C_2(j_2))|] \right).
\end{aligned}$$

**Step 4: Concentration bounds adapted to low-rank.** For  $Q_{2,1,k}$ , we have:

$$\begin{aligned} \max_{j_2 \leq n_2} \mathbb{E}_Z[Q_{2,1,k}(X, j_2)] &\leq K \max_{j_1 \leq n_1, 1 \leq k \leq r} |\tilde{w}_1(t, j_1, k) - w_1(t, C_1(j_1), k)| \\ &\leq K_t \mathcal{D}_t(W, \tilde{W}). \end{aligned}$$

For  $Q_{2,2,k}$ , we apply concentration. Let us write:

$$Z_{1,k}(x, c_1) = w_1(t, c_1, k) \varphi_1(\langle f_1(c_1), x \rangle).$$

Recall that  $\{C_1(j_1)\}_{j_1=1}^{n_1}$  are i.i.d. We have:

$$\mathbb{E}[Z_{1,k}(X, C_1(j_1)) | X] = \mathbb{E}_{C_1}[Z_{1,k}(X, C_1)],$$

and  $\{Z_{1,k}(X, C_1(j_1))\}_{j_1=1}^{n_1}$  are independent conditional on  $X$ . By Assumption ??,  $|Z_{1,k}(X, C_1(j_1))| \leq K_t$  almost surely. Then by concentration (similar to Lemma ??):

$$\mathbb{P}(\mathbb{E}_Z[Q_{2,2,k}(X, j_2)] \geq K_t \gamma_{2,k}) \leq \frac{1}{\gamma_{2,k}} \exp\left(-\frac{n_1 \gamma_{2,k}^2}{K_t}\right).$$

For  $Q_{B,2,k}$ , we note that almost surely:

$$|L_{C_2(j_2),k} \varphi'_2(H_2(t, C_2(j_2)); x, W) w_2(t, C_2(j_2))| \leq K |L_{C_2(j_2),k}| \leq K^2,$$

by Assumption ?. Then by concentration (similar to Lemma ??):

$$\mathbb{P}(\mathbb{E}_Z[Q_{B,2,k}(X)] \geq K_t \gamma_{1,k}) \leq \frac{1}{\gamma_{1,k}} \exp\left(-\frac{n_2 \gamma_{1,k}^2}{K_t}\right).$$

**Step 5: Combining bounds with union over  $k$ .** Taking a union bound over  $k = 1, \dots, r$  channels, we obtain that for any  $\gamma_1, \gamma_2 > 0$  and  $t \geq 0$ , the event:

$$\begin{aligned} &\max \left\{ \max_{j_2 \leq n_2} \mathbb{E}_Z[|q_2(t, X, j_2, C_2(j_2))|], \right. \\ &\quad \left. \max_{j_1 \leq n_1, 1 \leq k \leq r} \mathbb{E}_Z[|q_{B,k}(t, X)|] \right\} \\ &\leq K_t(1 + rK)(\mathcal{D}_t(W, \tilde{W}) + \gamma_1 + \gamma_2), \end{aligned}$$

holds with probability at least:

$$1 - \frac{n_1 r}{\gamma_1} \exp\left(-\frac{n_2 \gamma_1^2}{K_t(1 + rK)^2}\right) - \frac{n_2 r}{\gamma_2} \exp\left(-\frac{n_1 \gamma_2^2}{K_t(1 + rK)^2}\right).$$

**Step 6: Gronwall argument.** Combining the bounds and taking a union bound over a discrete time grid  $t \in \{0, \xi, 2\xi, \dots, \lfloor T/\xi \rfloor \xi\}$  for some  $\xi \in (0, 1)$ , we obtain:

$$\begin{aligned} & \max \left\{ \max_{j_2 \leq n_2} \left| \frac{\partial}{\partial t} \tilde{w}_2(t, j_2) - \frac{\partial}{\partial t} w_2(t, C_2(j_2)) \right|, \right. \\ & \quad \left. \max_{j_1 \leq n_1, 1 \leq k \leq r} \left| \frac{\partial}{\partial t} \tilde{w}_1(t, j_1, k) - \frac{\partial}{\partial t} w_1(t, C_1(j_1), k) \right| \right\} \\ & \leq K_T(1 + rK)(\mathcal{D}_t(W, \tilde{W}) + \gamma_1 + \gamma_2 + \xi), \quad \forall t \in [0, T], \end{aligned}$$

with probability at least:

$$1 - \frac{T+1}{\xi} \left[ \frac{n_1 r}{\gamma_1} \exp \left( -\frac{n_2 \gamma_1^2}{K_T(1+rK)^2} \right) + \frac{n_2 r}{\gamma_2} \exp \left( -\frac{n_1 \gamma_2^2}{K_T(1+rK)^2} \right) \right].$$

The above event implies:

$$\mathcal{D}_t(W, \tilde{W}) \leq K_T(1 + rK) \int_0^t (\mathcal{D}_s(W, \tilde{W}) + \gamma_1 + \gamma_2 + \xi) ds,$$

and hence by Gronwall's lemma and the fact  $\mathcal{D}_0(W, \tilde{W}) = 0$ :

$$\mathcal{D}_T(W, \tilde{W}) \leq (\gamma_1 + \gamma_2 + \xi) e^{K_T(1+rK)}.$$

The result follows from choosing:

$$\begin{aligned} \xi &= \frac{1}{\sqrt{n_{\max}}}, \quad \gamma_1 = \frac{K_T(1+rK)}{\sqrt{n_2}} \log^{1/2} \left( \frac{3(T+1)n_{\max}^2 r}{\delta} + e \right), \\ \gamma_2 &= \frac{K_T(1+rK)}{\sqrt{n_1}} \log^{1/2} \left( \frac{3(T+1)n_{\max}^2 r}{\delta} + e \right). \end{aligned}$$

□

## 1.5 Proof of Lemma 1.3

*Proof.* The proof follows the same structure as the full-width case [?], with careful adaptations for the low-rank structure. The key difference is that we need to account for the  $r$  channels in  $w_1$  and the mixing matrix  $L$  in all bounds.

We consider  $t \leq T$  for a given terminal time  $T \in \epsilon \mathbb{N}_{\geq 0}$ . We reuse the notation  $K_t$  from the proof of Lemma 1.3. Note that  $K_t \leq K_T$  for all  $t \leq T$ . We also note that at initialization,  $\mathcal{D}_0(\mathbf{W}, \tilde{W}) = 0$ .



For brevity, let us define quantities that relate to the difference in the gradient updates between  $\mathbf{W}$  and  $\tilde{W}$ :

$$\begin{aligned}
q_2(k, z, \tilde{z}, j_2) &= d_L(z; \mathbf{W}(k)) \varphi_2(\mathbf{H}_2(k, j_2; x, \mathbf{W}(k))) \\
&\quad - d_L(\tilde{z}; \tilde{W}(k\epsilon)) \varphi_2(\tilde{H}_2(k\epsilon, j_2; \tilde{x}, \tilde{W}(k\epsilon))), \\
r_2(k, z, j_2) &= \xi_2(k\epsilon) d_L(z; \tilde{W}(k\epsilon)) \varphi_2(\tilde{H}_2(k\epsilon, j_2; x, \tilde{W}(k\epsilon))) \\
&\quad - \xi_2(k\epsilon) \mathbb{E}_Z[d_L(Z; \tilde{W}(k\epsilon)) \varphi_2(\tilde{H}_2(k\epsilon, j_2; X, \tilde{W}(k\epsilon)))], \\
q_1(k, z, \tilde{z}, j_1, k') &= d_L(z; \mathbf{W}(k)) \varphi_1(\langle f_1(C_1(j_1)), x \rangle) \mathbf{B}_{k'}(k; x, \mathbf{W}(k)) \\
&\quad - d_L(\tilde{z}; \tilde{W}(k\epsilon)) \varphi_1(\langle f_1(C_1(j_1)), \tilde{x} \rangle) \tilde{B}_{k'}(k\epsilon; \tilde{x}, \tilde{W}(k\epsilon)), \\
r_1(k, z, j_1, k') &= \xi_1(k\epsilon) d_L(z; \tilde{W}(k\epsilon)) \varphi_1(\langle f_1(C_1(j_1)), x \rangle) \tilde{B}_{k'}(k\epsilon; x, \tilde{W}(k\epsilon)) \\
&\quad - \xi_1(k\epsilon) \mathbb{E}_Z[d_L(Z; \tilde{W}(k\epsilon)) \varphi_1(\langle f_1(C_1(j_1)), X \rangle) \tilde{B}_{k'}(k\epsilon; X, \tilde{W}(k\epsilon))],
\end{aligned}$$

where  $\mathbf{B}_{k'}(k; x, \mathbf{W}) = \frac{1}{n_2} \sum_{j_2=1}^{n_2} L_{C_2(j_2), k'} \varphi_2'(\mathbf{H}_2(k, j_2; x, \mathbf{W})) \mathbf{w}_2(k, j_2)$ .

By time-interpolation estimates (similar to Claim 1 in the full-width case) and Assumption ??, we have:

$$\begin{aligned}
|\mathbf{w}_2(\lfloor t/\epsilon \rfloor, j_2) - \tilde{w}_2(t, j_2)| &\leq K \max_{j_2 \leq n_2} [Q_{2,1}(\lfloor t/\epsilon \rfloor, j_2) + Q_{2,2}(\lfloor t/\epsilon \rfloor, j_2)] + tK_t\epsilon, \\
|\mathbf{w}_1(\lfloor t/\epsilon \rfloor, j_1, k) - \tilde{w}_1(t, j_1, k)| &\leq K \max_{j_1 \leq n_1, 1 \leq k \leq r} [Q_{1,1}(\lfloor t/\epsilon \rfloor, j_1, k) + Q_{1,2}(\lfloor t/\epsilon \rfloor, j_1, k)] + tK_t\epsilon,
\end{aligned}$$

where:

$$\begin{aligned}
Q_{2,1}(\lfloor t/\epsilon \rfloor, j_2) &= \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} |q_2(\ell, z(\ell), z(\ell), j_2)|, \\
Q_{2,2}(\lfloor t/\epsilon \rfloor, j_2) &= \left| \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} r_2(\ell, z(\ell), j_2) \right|, \\
Q_{1,1}(\lfloor t/\epsilon \rfloor, j_1, k) &= \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} |q_1(\ell, z(\ell), z(\ell), j_1, k)|, \\
Q_{1,2}(\lfloor t/\epsilon \rfloor, j_1, k) &= \left| \epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} r_1(\ell, z(\ell), j_1, k) \right|.
\end{aligned}$$

**Bounding the terms:** For  $Q_{2,1}$ , using Assumption ?? and the low-rank structure:

$$\begin{aligned}
|q_2(\ell, z(\ell), z(\ell), j_2)| &\leq K |d_L(z(\ell); \mathbf{W}(\ell)) - d_L(z(\ell); \tilde{W}(\ell\epsilon))| \\
&\quad + K |d_L(z(\ell); \tilde{W}(\ell\epsilon))| |\mathbf{H}_2(\ell, j_2; x(\ell), \mathbf{W}(\ell)) - \tilde{H}_2(\ell\epsilon, j_2; x(\ell), \tilde{W}(\ell\epsilon))| \\
&\leq K_t \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}) + K_t \sum_{k=1}^r |L_{C_2(j_2), k}| \max_{j_1 \leq n_1} |\mathbf{w}_1(\ell, j_1, k) - \tilde{w}_1(\ell\epsilon, j_1, k)| \\
&\leq K_t(1 + rK) \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}),
\end{aligned}$$

which yields:

$$\max_{j_2 \leq n_2} Q_{2,1}(\lfloor t/\epsilon \rfloor, j_2) \leq K_t(1+rK)\epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}).$$

For  $Q_{1,1}$ , similarly:

$$\begin{aligned} |q_1(\ell, z(\ell), z(\ell), j_1, k)| &\leq K|d_L(z(\ell); \mathbf{W}(\ell)) - d_L(z(\ell); \tilde{W}(\ell\epsilon))| |\mathbf{B}_k(\ell; x(\ell), \mathbf{W}(\ell))| \\ &\quad + K|d_L(z(\ell); \tilde{W}(\ell\epsilon))| |\mathbf{B}_k(\ell; x(\ell), \mathbf{W}(\ell)) - \tilde{B}_k(\ell\epsilon; x(\ell), \tilde{W}(\ell\epsilon))| \\ &\leq K_t(1+rK)\mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}), \end{aligned}$$

which yields:

$$\max_{j_1 \leq n_1, 1 \leq k \leq r} Q_{1,1}(\lfloor t/\epsilon \rfloor, j_1, k) \leq K_t(1+rK)\epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}).$$

For  $Q_{2,2}$  and  $Q_{1,2}$ , we use martingale concentration. The martingale differences are bounded:  $|r_2(\ell, z(\ell), j_2)| \leq K_t$  and  $|r_1(\ell, z(\ell), j_1, k)| \leq K_t(1+rK)$  almost surely by Assumption ?? and the low-rank structure. Then by martingale concentration (similar to Theorem ??):

$$\begin{aligned} \mathbb{P}\left(\max_{j_2 \leq n_2} \max_{\ell \in \{0,1,\dots,T/\epsilon\}} Q_{2,2}(\ell, j_2) \geq \xi\right) &\leq 2n_2 \exp\left(-\frac{\xi^2}{K_T(T+1)\epsilon}\right), \\ \mathbb{P}\left(\max_{j_1 \leq n_1, 1 \leq k \leq r} \max_{\ell \in \{0,1,\dots,T/\epsilon\}} Q_{1,2}(\ell, j_1, k) \geq \xi\right) &\leq 2n_1 r \exp\left(-\frac{\xi^2}{K_T(1+rK)^2(T+1)\epsilon}\right). \end{aligned}$$

**Putting everything together:** All the above results give us:

$$\mathcal{D}_{\lfloor t/\epsilon \rfloor \epsilon}(\tilde{W}, \mathbf{W}) \leq K_T(1+rK)\epsilon \sum_{\ell=0}^{\lfloor t/\epsilon \rfloor - 1} \mathcal{D}_{\ell\epsilon}(\tilde{W}, \mathbf{W}) + \xi + TK_T\epsilon, \quad \forall t \leq T,$$

which holds with probability at least:

$$1 - 2n_1 r \exp\left(-\frac{\xi^2}{K_T(1+rK)^2(T+1)\epsilon}\right) - 2n_2 \exp\left(-\frac{\xi^2}{K_T(T+1)\epsilon}\right).$$

The above event implies, by Gronwall's lemma:

$$\mathcal{D}_T(\tilde{W}, \mathbf{W}) \leq (\xi + \epsilon)e^{K_T(1+rK)}.$$

Choosing  $\xi = K_T(1+rK)\sqrt{(T+1)\epsilon \log(2n_1 n_2 r/\delta)}$  completes the proof.  $\square$

## 1.6 Discussion: Width Requirement of $\approx 1000$ Neurons

The bound in Theorem 1.1 shows that the approximation error scales as:

$$\text{err}_{\delta,T}(\epsilon, n_1, n_2, r) \leq e^{K_T(1+rK)} \left( \frac{1}{\sqrt{n_{\min}}} + \sqrt{\epsilon} \right) \log^{1/2} \left( \frac{3(T+1)n_{\max}^2}{\delta} + e \right).$$

For typical values  $T \leq 10$ ,  $K \approx 1$ ,  $r \leq 100$ , and  $\epsilon \approx 0.001$ , we have  $K_T(1+rK) \leq 1000$  (roughly), so  $e^{K_T(1+rK)} \leq e^{1000}$  (which is large but finite). The key term is  $1/\sqrt{n_{\min}}$ .

For  $n_{\min} = 1000$ , we have  $1/\sqrt{n_{\min}} = 0.032$ , which is small. Combined with  $\sqrt{\epsilon} \approx 0.032$  for  $\epsilon = 0.001$ , and the logarithmic factor  $\log^{1/2}(3(T+1)n_{\max}^2/\delta + e) \approx 10$  for  $n_{\max} = 1000$ ,  $T = 10$ ,  $\delta = 0.01$ , we get:

$$\text{err}_{\delta,T} \leq e^{1000} \times 0.064 \times 10 \approx (\text{manageable for practical purposes}).$$

More importantly, the bound shows that the error *decreases* as  $n_{\min}$  increases, and the rate  $O(1/\sqrt{n_{\min}})$  is independent of data dimension  $d$ . This explains why width  $\approx 1000$  is typically sufficient: the  $1/\sqrt{n_{\min}}$  term becomes small enough that mean-field behavior dominates the finite-width corrections.

Empirical validation in [?] confirms that networks with width  $\approx 1000$  trained on high-dimensional real-world data (e.g.,  $d \approx 1000$ ) exhibit mean-field behaviors, supporting our theoretical prediction that the required width is independent of  $d$ .